

MongoDB Mastery

A hands-on course in 7 sections and 10 labs, built on one university database and run on a free MongoDB Atlas cluster.

Every lab follows the same story, a student and course management system, so each new idea builds on data you already know. Each lab comes with a runnable notebook, a written walkthrough, an assignment with answer key, and automated tests. The course takes about 8 hours of lab time including a short, single-sitting capstone.

Difficulty key

Beginner	Intermediate	Inter. to Advanced	Advanced	Capstone
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The course at a glance

Lab	Title	Difficulty	Time	Requires
Lab 1	Student Records Lookup	Beginner	~35 min	None
Lab 2	Student Enrollment Tracker	Beginner	~35 min	Lab 1
Lab 3	Academic Performance Analytics	Intermediate	~40 min	Labs 1-2
Lab 3B	Query Optimization (new)	Intermediate	~50 min	Lab 3
Lab 4	Courses and Instructors	Intermediate	~40 min	Labs 1-3
Lab 4B	Schema Patterns and Anti-patterns (new)	Intermediate	~45 min	Lab 4
Lab 4C	Atlas Search and Vector Search (new)	Intermediate to Advanced	~60 min	Lab 4
Lab 5	Enrollment Transactions	Advanced	~40 min	Labs 1-4
Lab 6	Scaling the University Database	Advanced	~40 min	Labs 1-5
Lab 7	University Registration Database	Capstone	~90 min, one sitting	Labs 1-6 (+3B, 4B, 4C)

Labs 3B, 4B and 4C are new. They keep letter numbers so the original seven labs and their tests stay unchanged.

Section 1: Foundations

Document model, connecting to Atlas, reading data

Lab 1 - Student Records Lookup

MongoDB Basics: How to Query Data

Beginner

Time: ~35 min | **Requires:** None

What it offers

- Connect to a real MongoDB Atlas cluster from Python
- Insert documents and filter, sort and project results
- First taste of the aggregation pipeline

Section 2: Writing Data

Full CRUD lifecycle

Lab 2 - Student Enrollment Tracker

MongoDB Basics: How to Write, Update & Delete Data

Beginner

Time: ~35 min | **Requires:** Lab 1

What it offers

- Enroll and withdraw students with insert, update and delete operations
- Update fields in place and use upserts
- Complete the create, read, update, delete lifecycle

Section 3: Analytics and Performance

Aggregation, indexing and query tuning

Lab 3 - Academic Performance Analytics

MongoDB Intermediate: How to Analyze Data at Scale

Intermediate

Time: ~40 min | **Requires:** Labs 1-2

What it offers

- Average grades per course with \$group and \$avg
- Grade distributions with \$bucket
- Create an index and prove it works with .explain() (COLLSCAN to IXSCAN)

Lab 3B - Query Optimization (new)

MongoDB Intermediate: How to Make Queries Fast

Intermediate

Time: ~50 min | Requires: Lab 3

What it offers

- Diagnose slow queries on 100,000 records using explain executionStats
- Design compound indexes with the ESR rule and write covered queries
- Find unused indexes with \$indexStats and measure what indexes cost
- Tune aggregations and finish with a before/after scoreboard

Section 4: Data Modeling

Schema design, patterns and anti-patterns

Lab 4 - Courses and Instructors

MongoDB Intermediate: How to Design Schemas & Search Data

Intermediate

Time: ~40 min | Requires: Labs 1-3

What it offers

- Decide between embedding and referencing related data
- Join collections with \$lookup
- Add basic text search over course descriptions

Lab 4B - Schema Patterns and Anti-patterns (new)

MongoDB Intermediate: How to Recognize Schema Patterns and Anti-patterns

Intermediate

Time: ~45 min | Requires: Lab 4

What it offers

- Spot the unbounded-array anti-pattern and the 16 MB document limit
- Apply the subset, computed, bucket and extended-reference patterns
- Measure document sizes and see the trade-offs in numbers

Section 5: Search and AI

Keyword search, vector search and hybrid ranking

Lab 4C - Atlas Search and Vector Search (new)

MongoDB Intermediate: How to Search by Keywords and by Meaning

Intermediate to
Advanced

Time: ~60 min | Requires: Lab 4

What it offers

- Keyword search with \$search: relevance scores, typo tolerance, filters
- Create embeddings locally (no API key) and search by meaning with \$vectorSearch
- Merge both result lists with Reciprocal Rank Fusion (hybrid search)
- Build the retrieval context block for a RAG prompt

Section 6: Reliability and Scale

Transactions, replication and security

Lab 5 - Enrollment Transactions

MongoDB Advanced: How to Guarantee Data Consistency

Advanced

Time: ~40 min | **Requires:** Labs 1-4

What it offers

- Multi-document ACID transactions: enroll a student and reduce seats atomically
- React to changes with change streams
- Back up and restore data

Lab 6 - Scaling the University Database

MongoDB Advanced: How to Inspect a Replica Set and Secure Access

Advanced

Time: ~40 min | **Requires:** Labs 1-5

What it offers

- Inspect a replica set and understand high availability
- Secure access with role-based access control and TLS
- Sharding explained conceptually (not possible on the free tier)

Section 7: Capstone

One short project that reuses every earlier lab

Lab 7 - University Registration Database

MongoDB Capstone: Build the Database Behind a Course Registration System

Capstone

Time: ~90 min, one sitting | **Requires:** Labs 1-6 (+3B, 4B, 4C)

What it offers

- Design, seed and maintain a registration database (CRUD, schema patterns)
- Enroll students with a transaction that can never over-book a course
- Report, tune and search the data: aggregation, ESR index, Atlas Search
- Starter and solution notebooks with self-checking cells

Learning path summary

Section	Labs	Focus	Time
1 Foundations	Lab 1	Connect, insert, query	~35 min
2 Writing Data	Lab 2	Create, update, delete	~35 min
3 Analytics and Performance	Labs 3, 3B	Aggregation, indexing, tuning	~90 min
4 Data Modeling	Labs 4, 4B	Schema design and patterns	~85 min
5 Search and AI	Lab 4C	Keyword and vector search	~60 min
6 Reliability and Scale	Labs 5, 6	Transactions, replicas, security	~80 min
7 Capstone	Lab 7	Registration database project	~90 min
Total	10 labs		about 8 hrs

How to use the course: follow the sections in order. Inside a section, do the numbered lab first and its lettered lab second. The Lab 7 capstone reuses one skill from every section, so do it last. Complete each assignment before checking its answer key. All labs run on the free Atlas M0 tier, except where noted (the profiler in Lab 3B and sharding in Lab 6 need a paid tier).